

Exercise: Implementing Tetris Game with Tkinter

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1 Objective

In this exercise, you must implement a simple version of the Tetris game using **Tkinter** in **Python**. The program should have a **graphical user interface** where the player can control the Tetris blocks and earn points.

2 Exercise Description

2.1 Creating the Game Window

- Use **Tkinter** to create a **graphical window**.
- It is recommended to create a **Canvas** to display the game board.

2.2 Defining Tetris Blocks

- Implement standard Tetris shapes (**I, O, T, L, J, S, Z**) as **2D matrices**.
- Display blocks using **different colors or custom images**.¹

2.3 Moving the Blocks

- Allow blocks to **move left, right, and down**.
- Enable **rotation** of blocks using the **Up** key.

2.4 Checking for Full Rows

- If a row is fully occupied, it should be removed.
- The rows above should **shift down** by one cell.

¹Using colors or images for blocks is **mandatory**.

2.5 Game Over Condition

- If new blocks cannot fit in the available space, the game should stop.
- Display the message “**Game Over**”.

2.6 Scoring System

- Increase score when a row is cleared.
- Display the score in a corner of the game window.

2.7 Timer and Automatic Falling

- Blocks should **automatically fall** by one cell every few milliseconds.
- The player can **speed up the fall** by pressing the **Down** key.

2.8 Object-Oriented Programming (OOP) Requirement

- The game must be implemented using **Object-Oriented Programming (OOP)** principles.
- Create appropriate **classes** for the game components.
- Use methods for handling game logic and interactions.

2.9 Game Levels and Difficulty Increase

- Display **game levels** and increase speed at each level. This feature is **mandatory**.

Good luck & Happy New Year!